

```
: "I will walk 500 miles and I would walk 500 more. Just to be the thousand  
: "I played the play playfully as the players were playing in the field."
```

```
import nltk  
nltk.download('punkt')  
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data]   Package punkt is already up-to-date!  
[nltk_data] Downloading package punkt_tab to /root/nltk_data...  
[nltk_data]   Package punkt_tab is already up-to-date!  
True
```

```
from nltk import word_tokenize, sent_tokenize  
  
print("-----Tokenized Words-----")  
print('Tokenized words:', word_tokenize(sentence1))  
print("\n")
```

```
-----Tokenized Words-----  
Tokenized words: ['I', 'will', 'walk', '500', 'miles', 'and', 'I', 'would', 'wal
```

```
print("-----Tokenized Sentences-----")  
print('Tokenized sentences:', sent_tokenize(sentence1))  
print("\n")
```

```
-----Tokenized Sentences-----  
Tokenized sentences: ['I will walk 500 miles and I would walk 500 more.', 'Just
```

```
import nltk  
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to  
[nltk_data]   /root/nltk_data...  
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.  
True
```

```
from nltk import pos_tag  
print("-----POS Tagging-----")  
token = word_tokenize(sentence1) + word_tokenize(sentence2)
```

```
print('POS tagged:', pos_tag(token))
print("\n")
```

```
-----POS Tagging-----
POS tagged: [('I', 'PRP'), ('will', 'MD'), ('walk', 'VB'), ('500', 'CD'), ('mile
```

```
import nltk
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
print("-----Stop-Words Removal-----")
from nltk.corpus import stopwords
stop_words = set(stopwords.words('english'))
token = word_tokenize(sentence1)
cleaned_token = []
for word in token:
    if word not in stop_words:
        cleaned_token.append(word)
print("Unclean version:", token)
print("\n")
print("Cleaned version:", cleaned_token)
print("\n")
```

```
-----Stop-Words Removal-----
Unclean version: ['I', 'will', 'walk', '500', 'miles', 'and', 'I', 'would', 'wal

Cleaned version: ['I', 'walk', '500', 'miles', 'I', 'would', 'walk', '500', '.',
```

```
print("-----Stemming-----")
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
token = word_tokenize(sentence2)
stemmed = [stemmer.stem(word) for word in token]
print("Stemmed words:", stemmed)
print("\n")
```

```
-----Stemming-----
Stemmed words: ['i', 'play', 'the', 'play', 'play', 'as', 'the', 'player', 'were
```

```
import nltk
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
True
```

```
print("-----Lemmatization-----")
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
token = word_tokenize(sentence2)
lemmatized = [lemmatizer.lemmatize(word) for word in token]
print("Lemmatized words:", lemmatized)
print("\n")
```

```
-----Lemmatization-----
Lemmatized words: ['I', 'played', 'the', 'play', 'playfully', 'a', 'the', 'playe
```

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